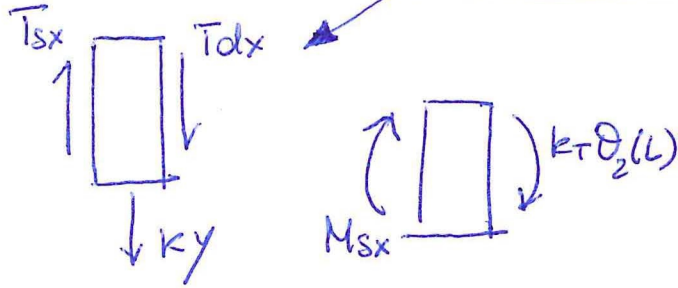
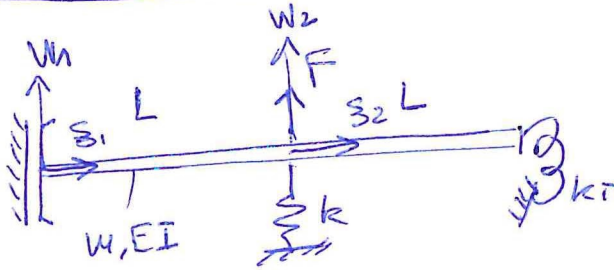


• PROBLEM 2



$$\begin{aligned} \theta_1(0) &= w_1'(0) = 0 \\ T_1(0) &= ET w_1'''(0) = 0 \\ w_1(L) &= w_2(0) \\ \theta_1(L) &= \theta_2(0) \rightarrow w_1'(L) = w_2'(0) \\ M_1(L) &= M_2(0) \rightarrow EI w_1''(L) = EI w_2''(0) \end{aligned}$$

$$\begin{aligned} T_{1x} - T_{2x} - ky &= 0 \rightarrow \\ \rightarrow EI w_1'''(L) - EI w_2'''(0) - k w_2(0) &= 0 \\ M_{1x} + k_T \theta_2(L) &= 0 \rightarrow \\ \rightarrow EI w_2''(L) + k_T w_2'(L) &= 0 \\ T_2(L) &= EI w_2'''(L) = 0 \end{aligned}$$

$$w_i(s_i) = A_i \cos \gamma s_i + B_i \sinh \gamma s_i + C_i \cosh \gamma s_i + D_i \sin \gamma s_i \quad i = 1, 2$$

$$\begin{aligned} B_1 + D_1 &= 0 \\ -B_1 + D_1 &= 0 \\ A_1 \cos \gamma L + B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sin \gamma L - A_2 - C_2 &= 0 \\ -A_1 \sinh \gamma L + B_1 \cos \gamma L + C_1 \sin \gamma L + D_1 \cosh \gamma L - B_2 - D_2 &= 0 \\ -A_1 \cos \gamma L - B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sin \gamma L + A_2 - C_2 &= 0 \\ (A_1 \sinh \gamma L - B_1 \cos \gamma L + C_1 \sin \gamma L + D_1 \cosh \gamma L) \gamma^3 EI - (-B_2 + D_2) \gamma^3 EI + \\ &\quad -k(A_2 + C_2) = 0 \\ (-A_2 \cos \gamma L - B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sin \gamma L) \gamma^2 EI + \\ &\quad + \gamma (-A_2 \sinh \gamma L + B_2 \cos \gamma L + C_2 \sin \gamma L + D_2 \cosh \gamma L) k_T = 0 \\ A_2 \sinh \gamma L - B_2 \cos \gamma L + C_2 \sin \gamma L + D_2 \cosh \gamma L &= 0 \end{aligned}$$

A ₁	B ₁	C ₁	D ₁	A ₂	B ₂	C ₂	D ₂
	1		1				
	-1		1				
cos γ L	seu γ L	Ch γ L	Sh γ L	-1		-1	
-seu γ L	cos γ L	Sh γ L	Ch γ L		-1		-1
-cos γ L	-seu γ L	Ch γ L	Sh γ L	1		-1	
seu γ L	-cos γ L	Sh γ L	Ch γ L	-cc	1	-cc	-1
				-cos γ L -pp seu γ L	-seu γ L +pp cos γ L	Ch γ L + +pp Sh γ L	Sh γ L + +pp Ch γ L
				seu γ L	-cos γ L	Sh γ L	Ch γ L

$$cc = \frac{k}{\gamma^3 EI} \quad \gamma \quad pp = \frac{k\tau}{\gamma EI}$$

$$V = V_{beam} + V_{LIN. SPR.} + V_{TRANS. SPR.} = V_{beam} + \frac{1}{2} k W_2^2(q, t) + \frac{1}{2} k\tau \theta_2^2(L, t) \rightarrow$$

$$\rightarrow K_{ui} = K_{u, beam} + k \phi_i^2(L) + k\tau \phi_i^2(2L) \quad i=1, N$$

$$\text{where } W(s, t) = \underline{\phi^T(s)} \underline{q(t)}$$

FRF : input and output in $s=L \rightarrow$

$$\rightarrow Q_{ui} = F_0 e^{i\omega t} \phi_i(L)$$

$$W(L, t) = \sum_{i=1}^N \phi_i(L) q_i(t)$$